



Minsky Lab Digital-Twin Junior

TOGA NETWORKS LTD., A HUAWEI COMPANY



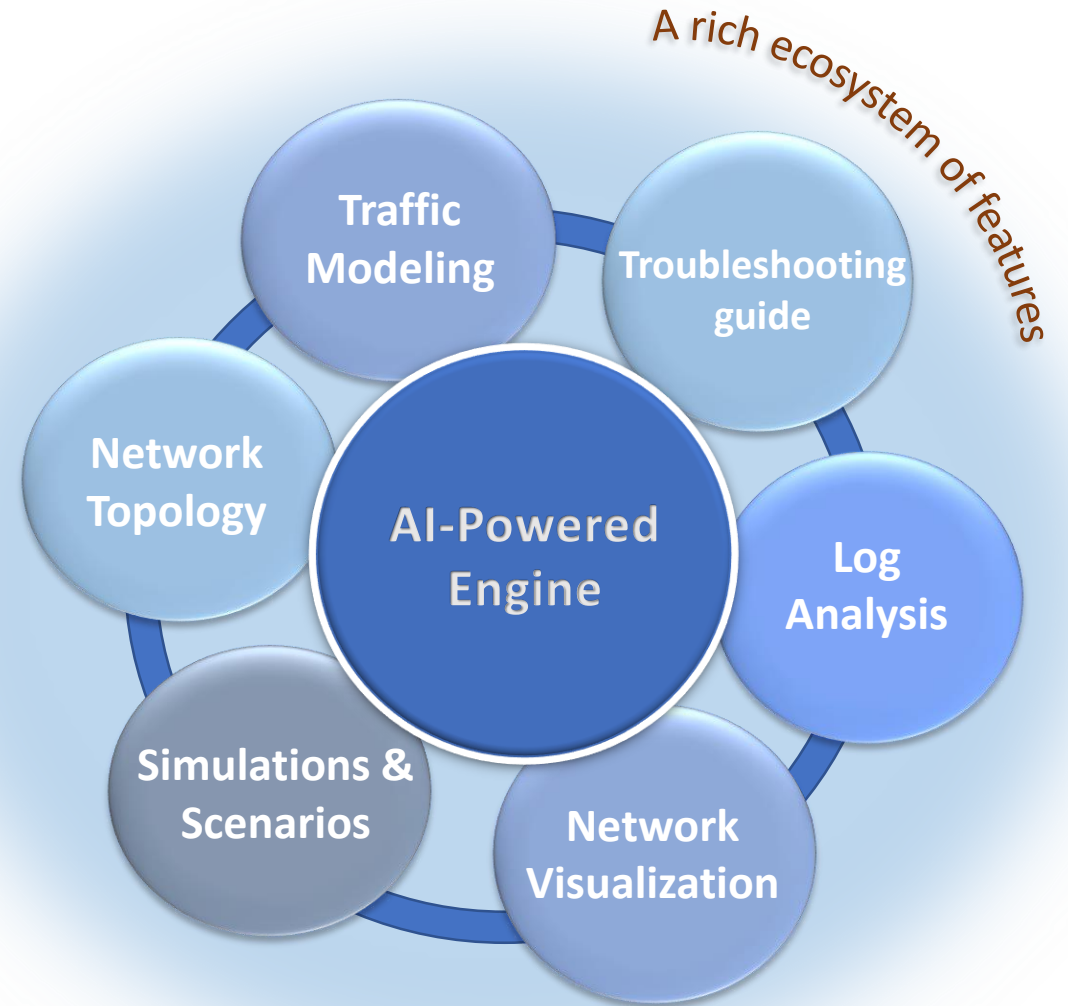
Digital Twin Junior:

a Comprehensive Portfolio
of AI-driven Network Intelligence Tools



Included capabilities:

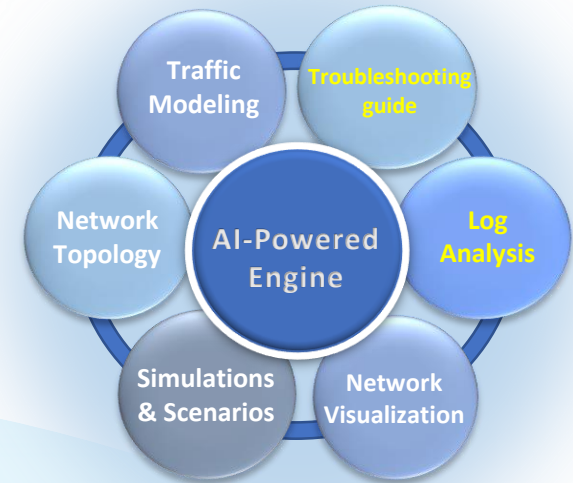
- ✓ Network simulation and scenario modeling
- ✓ Traffic Modeling: analysis and forecasting
- ✓ Network Topology mapping and diagnostics
- ✓ Automated log interpretation



+ **Interactive tools:** Network visualization, Troubleshooting guide

Digital Twin Junior:

Log Analysis and Interactive Troubleshooting



Log entry explanation

Network status extraction: logs, alarms

```
2019-06-12T10:20:30.000Z WARNING
PKI/4/HTTP_GET_CRL_ERR: file_name:[test.crl] Manually
obtaining CRL [test.crl] through HTTP failed (Reason=[URL
resolution failed]). category=PKI subCategory=CRL
component=HTTP,
eventID=HTTP_GET_CRL_ERR,vendor=Huawei,product=Fat
AP,version=V200R022C00, timestamp: [2019-06-12 10:20:30] [
rid:T6zHJ38AAAEAAo2BCwAAAMk
ip:[192.158.1.38]:1813/UDP
```

```
2019-06-13T11:21:31.000Z WARNING
PKI/4/HTTP_GET_CRL_ERR: file_name:[test1.crl] Manually
obtaining CRL [test1.crl] through HTTP failed (Reason: Failed to
save the response message). category=PKI subCategory=CRL
eventID=HTTP_GET_CRL_ERR,vendor=Huawei,product=Fat
AP,version=V200R022C00, timestamp: [2019-06-13 11:21:31] [
rid:T6zHJ38AAAEAAo2BCwAAAMk
ip:[192.158.1.39]:1813/UDP
```

Alarm Description:

After receiving a Hello packet from a neighbor, the device records information about the neighbor and starts a timer for the neighbor. If the device does not receive any Hello packet from the PIM neighbor before the timer expires, the device reports a trap.

Log Name	Describes the log name)	hwPimNeighborLoss
NbrIntIndex	Index of the interface where the neighbor resides	2
NbrUpTime	Time when a neighbor relationship is established	2020-12-12 10:20:30

Recommendation for a solution

1. Check whether the reset msdp peer command is run in the user view on the peer device.
 - If is performed, wait until the peer is re-established.
 - If not, go to Step 2.
2. Run the display this command in the MSDP view of the peer device to check whether an MSDP peer is configured.
 - If yes, go to step 3.
 - If no, run the peer connect-interface <interface-type> <interface-number> command in the MSDP view to configure an MSDP peer.
3. Run the display interface brief command to check whether the shutdown command is configured on the peer interface.
 - If so, run the undo shutdown command in the interface view to clear the alarm.
 - If no, go to step 4.
4. Collect alarm and configuration information, and contact technical support personnel.

Summary of the incident

The root cause of hwPimNeighborLoss was investigated. Here are the main findings:
Step 1: Ping/Ping IPv6 command outcome: broken (link between neighbors is not normal)
Step 2: Display interface brief/Display IPv6 interface brief command outcome: no, mistake (link is normal)
Step 3: Display IP routing-table/Display IPv6 routing-table command outcome: seems ok
Conclusion: The problem was not solved.

Interactive step-by-step procedure following

Chatbot

broken

Next Instruction: Run the display interface brief or display ipv6 interface brief command to check the interface status and check whether the protocol status is Up. Please provide the outcome

no, mistake, the link is normal!

Next Instruction: Run the display ip routing-table or display ipv6 routing-table command to check whether the unicast route to the neighbor is normal. Please provide the outcome

- Identify API call
- Extract arguments
- Execute
- Get output

Key abilities of the network assistant

- ✓ Analyze and explain network events
- ✓ Identify causes
- ✓ Suggest solutions
- ✓ Resolve alarms
- ✓ Summarize incidents

Digital Twin Junior:

Monitoring and Analysis solution

1. **Novel statistical model for reliability evaluation:** Simulates network reliability with Monte-Carlo precision, factoring in aging and failures.
2. **Dynamic traffic-model for long-term analysis:** Adapts to traffic trends and load distribution, optimizing network performance.
3. **Multi-objective performance analysis:** Evaluates multiple metrics simultaneously, covering both physical and logical network aspects.
4. **Quasi-real time performance and high scalability:** Calculates performance in seconds for 1,000+ networks, and in minutes for up to 500k networks.
5. **“One Click” modification of network topology:** Instantly assess the impact of network changes with easy configuration tweaks.
6. **Comparison analysis of multiple scenarios:** Compares and recommends the best network architectures and configurations.
7. **High accuracy:** Achieves 98-99% reliability accuracy, outperforming traditional methods.
8. **Interactive Network Visualization Tool:** Offers a clear, detailed layout of network layers, interconnections, and component information for easy management.

Datacom Gold Network issued by Backbone Router Design Group (Module)



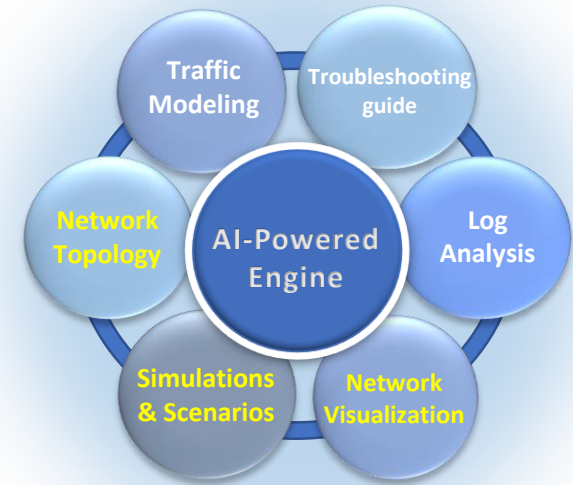
主笔寄语

TO: 柯西Lab可靠性建模和算法组-Cauchy Lab Reliability Modeling and Algorithms Group

获奖成员: Pavel Kisilev/408278;董峰/444526;Felix Goldberg/541990;Shaked Leibziner/650385;饶家鼎/594886;甄朋/468867;陈牧歌/623354;Leon Bruckman/523644;Ilan Bar/507596

柯西Lab基于蒙特卡洛模拟的可靠性算法可实现3K左右节点分钟级的仿真，相对于传统算法如RBD,Markov过程等小时级的运算时间性能大幅提升，并且可支持任意差异化组网、差异化业务目标，满足了骨干路由器框式和交换机盒式组网的可靠性量化对比诉求，可用于后续骨干路由器的网络架构设计，客户组网和扩容引导等场景。Cauchy Lab's reliability algorithm based on Monte Carlo simulation can achieve minute-level simulation at about 3K nodes. Compared with traditional algorithms such as RBD and Markov process, the hour-level computing time performance is greatly improved. In addition, it can support differentiated networking schemes and differentiated service objectives. Meets the requirement of comparing frame routers and box switches. It can be used in scenarios such as network architecture design of backbone routers, customer networking and capacity expansion guidance.

FROM: 骨干路由器设计组（模块）



Our own developed Apparatus for Multi-objective network performance comparison was accepted as a new Huawei patent

Invention Evaluation Approved (Invention ID: 92036486)

75435 14:13

Dear Sir/Madam,

Please be advised that the invention submitted has been approved for patent filing. Please log in Patent Management System to complete subsequent tasks if any.

Invention ID:92036486

Title:Apparatus for Multi-objective network performance comparison

Our own developed SmartFlow algorithm for Real-time Load Distribution for High-Performance Networks was recognized as a Huawei “business secret” patent

**Digital Twin Junior**
HUAWEI DATA COM AI LAB, TRC

Load Network

Choose File No file chosen

Load Traffic Trends

Choose File No file chosen

Modify Network

Deactivate

MTBF Of New Component [yrs]
Please Insert A Positive Value

MTTR Of New Component [hrs]
Please Insert A Positive Value

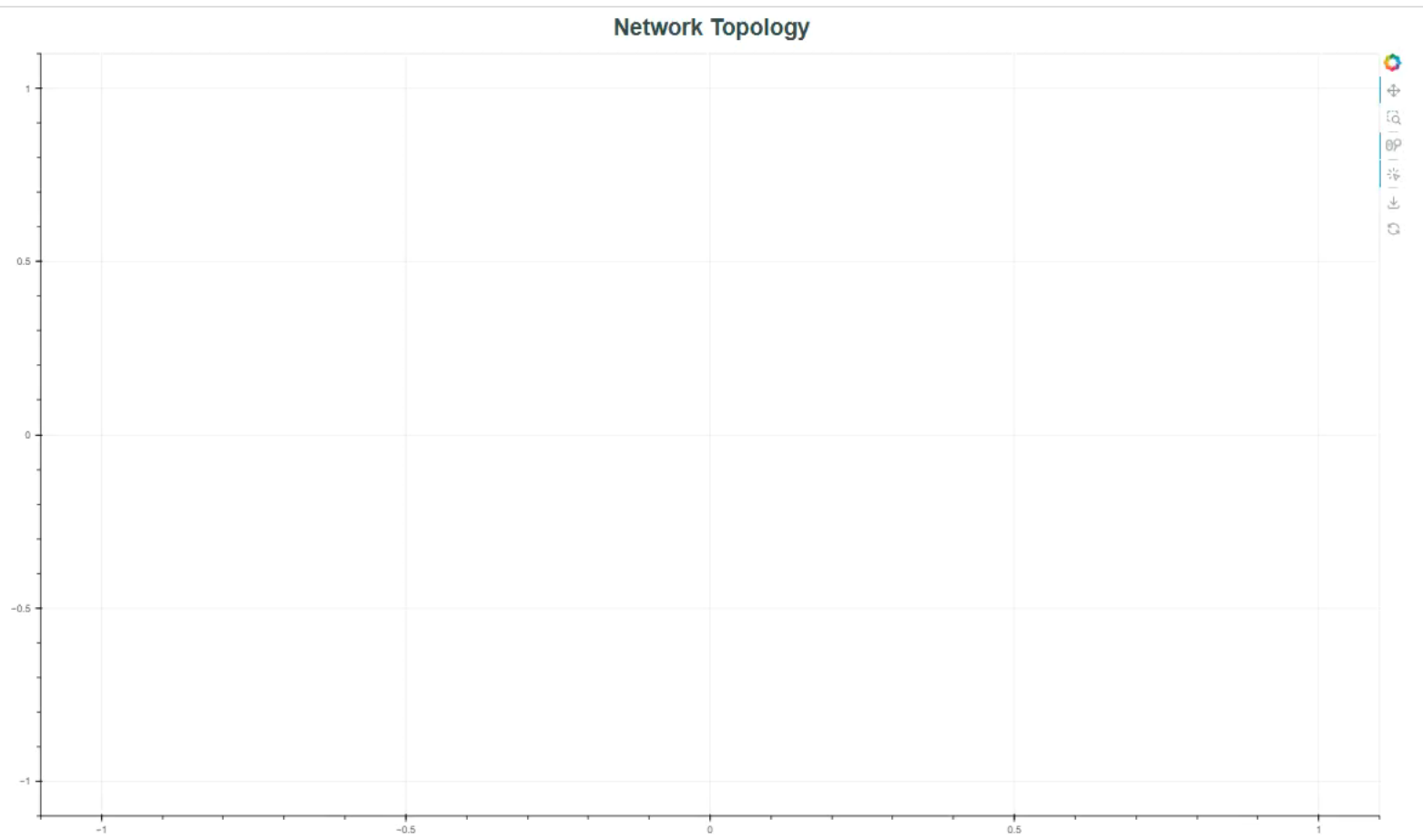
Link Multiplicity [#]
Please Insert An Integer

Execute Modification

Run Network Analysis

Fix Demand Compliance

Reset



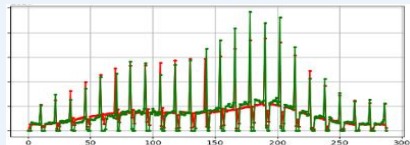
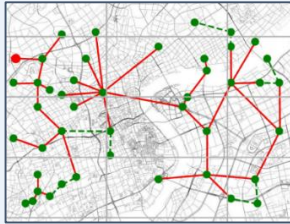
Action Description:

PERFORMANCE ANALYSIS		TRAFFIC STATISTICS	SLA AWARENESS
Metric	Result		
Convergence Time			

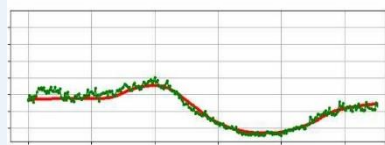
Digital Twin Junior: Traffic Modeling

Forecasting

- Horizon – from very short (1 min) to very long (1 day)
- Predict peaks & ‘quite periods’ – be prepared for overloading and energy saving time
- Unstable signal (e.g. hardly responding cell towers) – no problem, prediction is based on neighborhood data
- Holidays and other key events – we learn and predict even unusual traffic patterns
- Different topologies – known or unknown



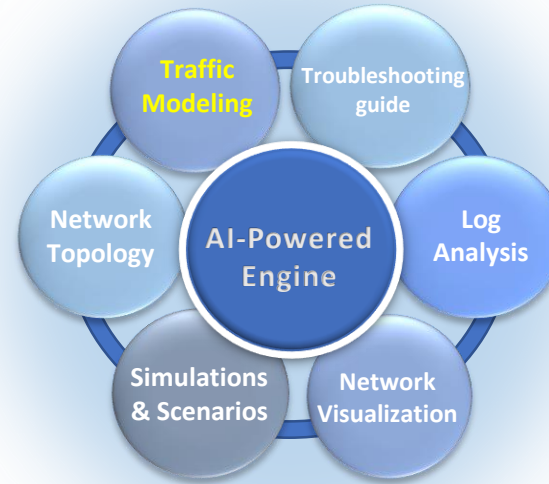
prediction restores peaks when they are signal



prediction ignores peaks when they are noise

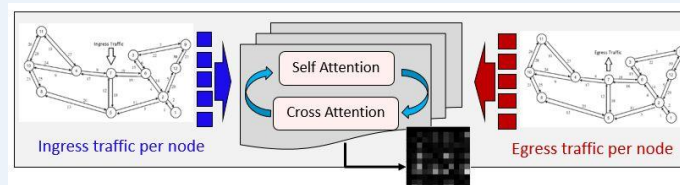
— ground truth

— prediction



Estimation

- Traffic Matrix from link loads: Route-Aware GNN
- Traffic Matrix from Ingress/Egress Traffic: Attention-based GNN (Deep Gravity)



Completion

Completion in Time

- ‘Estimate the present’ – fill in the missing values (traffic imputation)
- ‘Predict the future’ almost without knowing the past (with up to 80% of the historical data missing)

Completion in Space

- Traffic Inversion for sampled traffic in network monitoring tools

Key techniques: Attention-based Graph Convolutional Neural Networks

Can be applied to: Backbone networks, Campus network, Datacenter networks

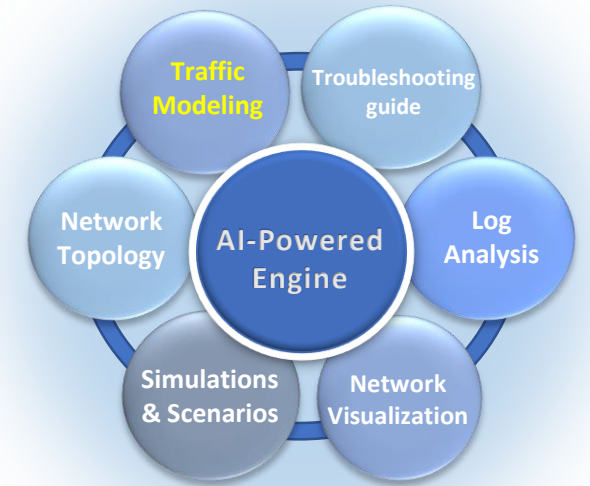
Digital Twin Junior: Traffic Inversion

Network monitoring

As networks evolve and become more complex, the need for network monitoring tools has remained constant.

It is highly important to ensure the network sustainability and guarantee all the processes to run smoothly and error-free for users.

Network traffic monitoring contains tools for data collection, analysis, diagnosis and solutions to resolve problematic issues.



Network Traffic Collection

The collection of **full detailed traffic** data does not scale well with increasing network attributes, like size, link bandwidth, speed etc.



Network Traffic Sampling

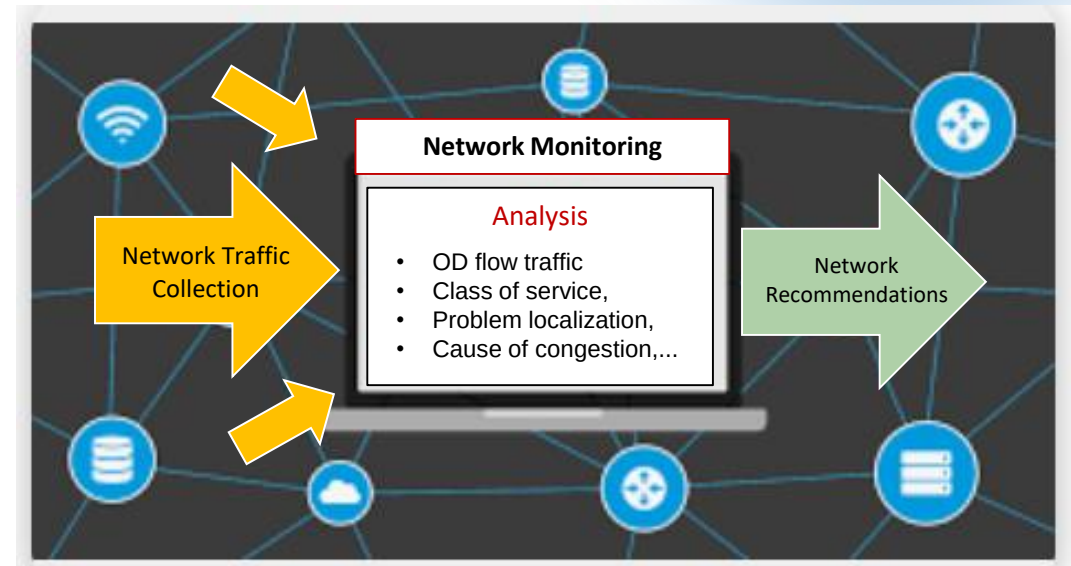
- The load on the measurement infrastructure is reduced.
- Problems: Only a portion of the traffic is exported. Some flows can be missed or represented by a very small portion. Sampled flows may not provide a sufficient amount of information about full network traffic



Network Traffic Inversion

Flow inversion is a restoration of the full traffic data or its statistics.

Existing approaches cannot restore the actual fine-grained traffic with good accuracy.



Some standards:

External products:

- NetFlow
- sFlow
- IPFIX

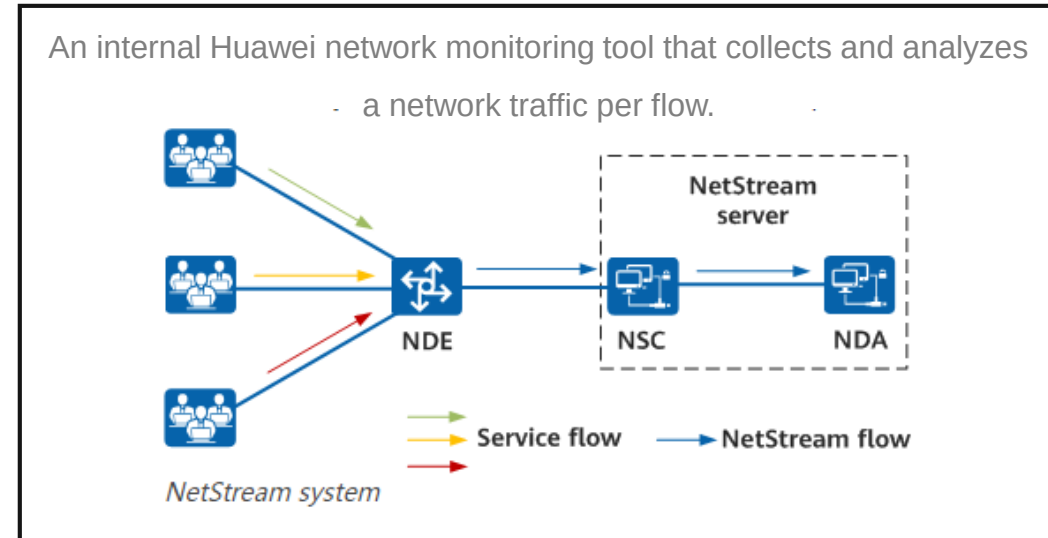
Huawei product:

- NetStream
(our main use case):

Full fine-grained traffic inversion

- In most cases sampling is inevitable...
- How to obtain a full information from sampled data?

NetStream



Full flows are not directly available from NetStream sampled data.

Let's consider 2 possibilities:



Statistics:

- ✓ Easy to calculate
- ✗ Loose a temporal structure of the flow data



DNN/AI tools:

- ✓ May be very accurate
- ✗ Require a lot of labeled data for the training



Input data

Device (router)
equipped with a network
monitoring tool



Network
Monitoring
Functionality



Standard sampled data tuple:

timestamp, {source IP, destination IP, input port, output port, PPS, BPS}

NetStream data example

	A	B	C	D	E	F	G	H
1	bps	deviceip	dstip	inputDeviceInterface	outputDeviceInterface	pps	srcip	timestamp
2	499600	10.79.253.46	10.58.128.98	14	15	50	10.243.33.3	1658688600
3	98600	10.79.253.46	7.186.131.9	79	14	110	10.58.83.203	1658688060

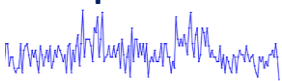
- ✓ Timestamp
- ✓ Source IP
- ✓ Destination IP
- ✓ Input port
- ✓ Output port

Information for traffic data pre-processing

- Subnet
- Application
- Group of servers
- Traffic clustering/detection per user

- ✓ PPS
- ✓ BPS

Sampled traffic



restore



Full traffic



- For specific end-user, subnet
- Total Ingress/Egress traffic inversion
- For application/group of applications

Full fine-grained traffic inversion

Data discussion:

- Lots of observations are available from monitoring devices  train DNN
- **Big problem:** our sampled data is not labeled (corresponding full traffic is not usually available)
- **Literature suggests:** incorporating knowledge of different noise and sampling patterns can significantly improve signal reconstruction
- So, to achieve accurate recovery, **it is crucial to replicate the sampling procedure** used by the monitoring device
- **Why not to just multiply by sampling ratio?** Even if total traffic is sampled deterministically (our use case) by inspecting every Rth packet, it would be generally inaccurate to assume that also every Rth application packet is considered

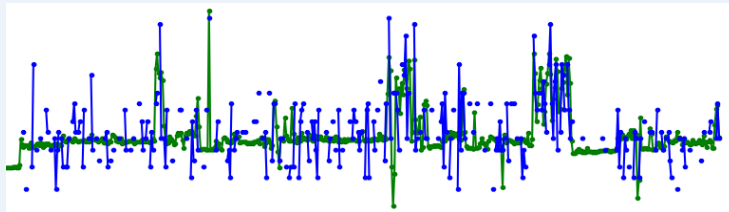
Our algorithm highlights

- ✓ Estimates an **actual time-dependent fine-grained traffic** with low error
- ✓ Combines statistical approach with recent AI techniques:
 - Is data-driven and AI-based: it reflects the actual data features
 - **Doesn't require labeled data**
 - Can be automatically adapted to a new data
- ✓ Is implemented as a lightweight and efficient deep neural network

Simple statistical manipulations

Assumption: if sampling ratio is know, we can scale back the sampled signal and obtain a good solution

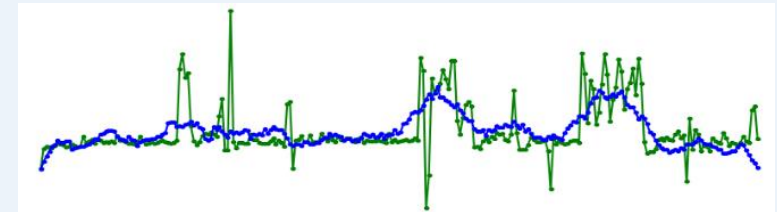
Compare true signal to scaled sampled signal



MAPE = 31%

— Sampling Ratio * sampled application traffic
— full application traffic

Compare true signal to **smoothed** scaled sampled signal



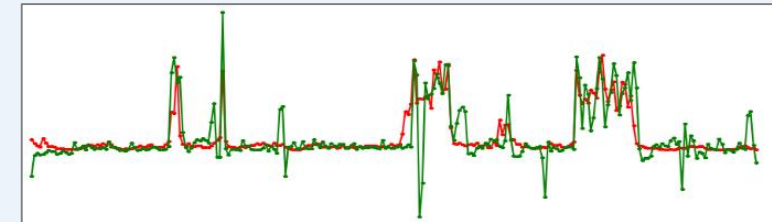
MAPE = 18%

— Sampling Ratio * smoothed sampled application traffic
— full application traffic

Simple statistics are not enough:

- ✓ **Good average** fit for a large time period
- ✗ **Poor local** fit per timestamp (either too noisy or oversmoothed)

Our model: Statistics + Attention-based DNN



MAPE = 12%

— Our result
— full application traffic

Train data modeling

Definitions:

R - sampling ratio

n - sampled number of packets

N - total number of packets

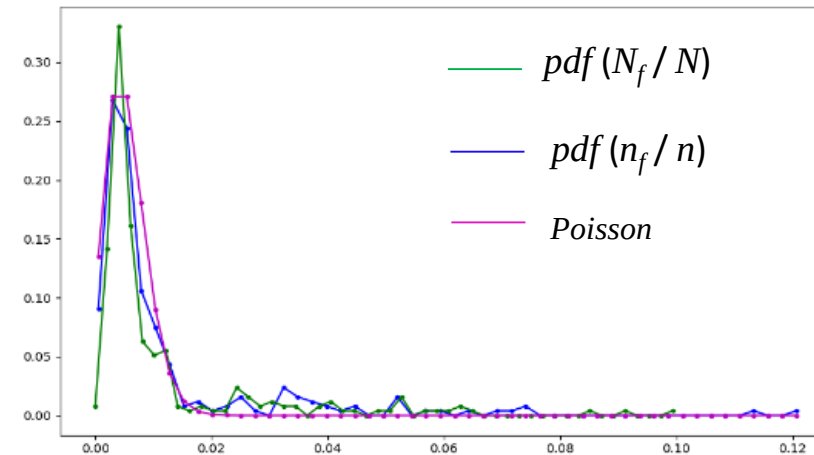
n_f - sampled number of flow packets (per application per user)

N_f - total number of flow packets

(traffic is aggregated)

Empirical result as a base for the sampling principle: $pdf(n_f / n) = pdf(N_f / N)$

$N_f \ll N$, hence the 'the law of small numbers' (Poisson) is straightforward to apply

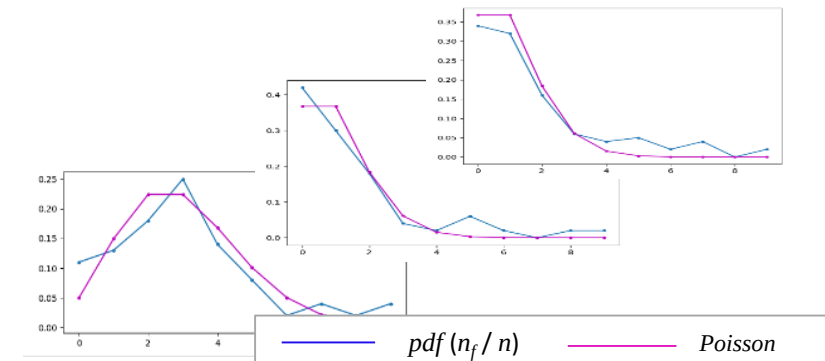


$pdf(n_f / n) \sim pdf(N_f / N) \sim \text{Poisson distribution}$

Approach:

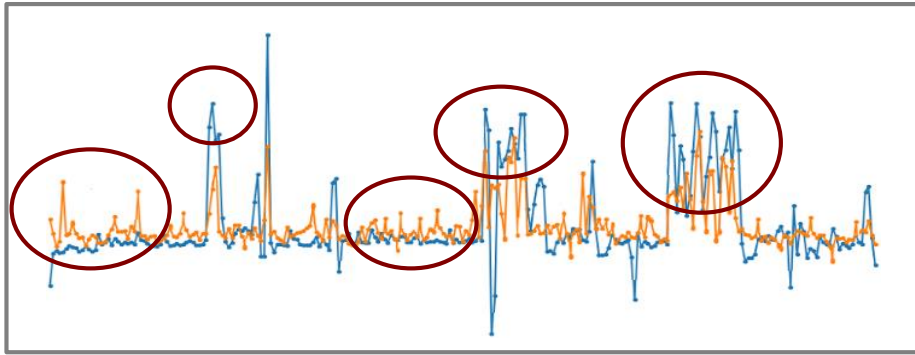
- Extract and apply a noise scheme/sampling to NetStream data to create a 'noisier' signal.
- Train a network to restore a noise signal from 'noisier' signal

Generated train data: inputs = n_f^{sampled} , labels = n_f^{original}

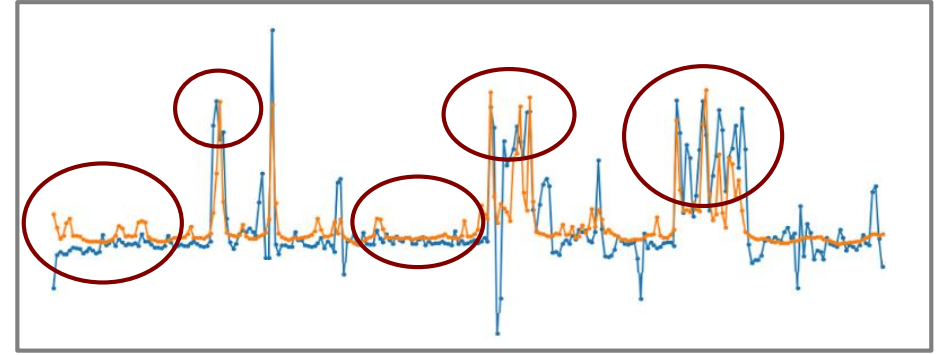


Spoiler: Why Poisson data modeling is better

Restored traffic (orange) vs ground truth traffic (blue)



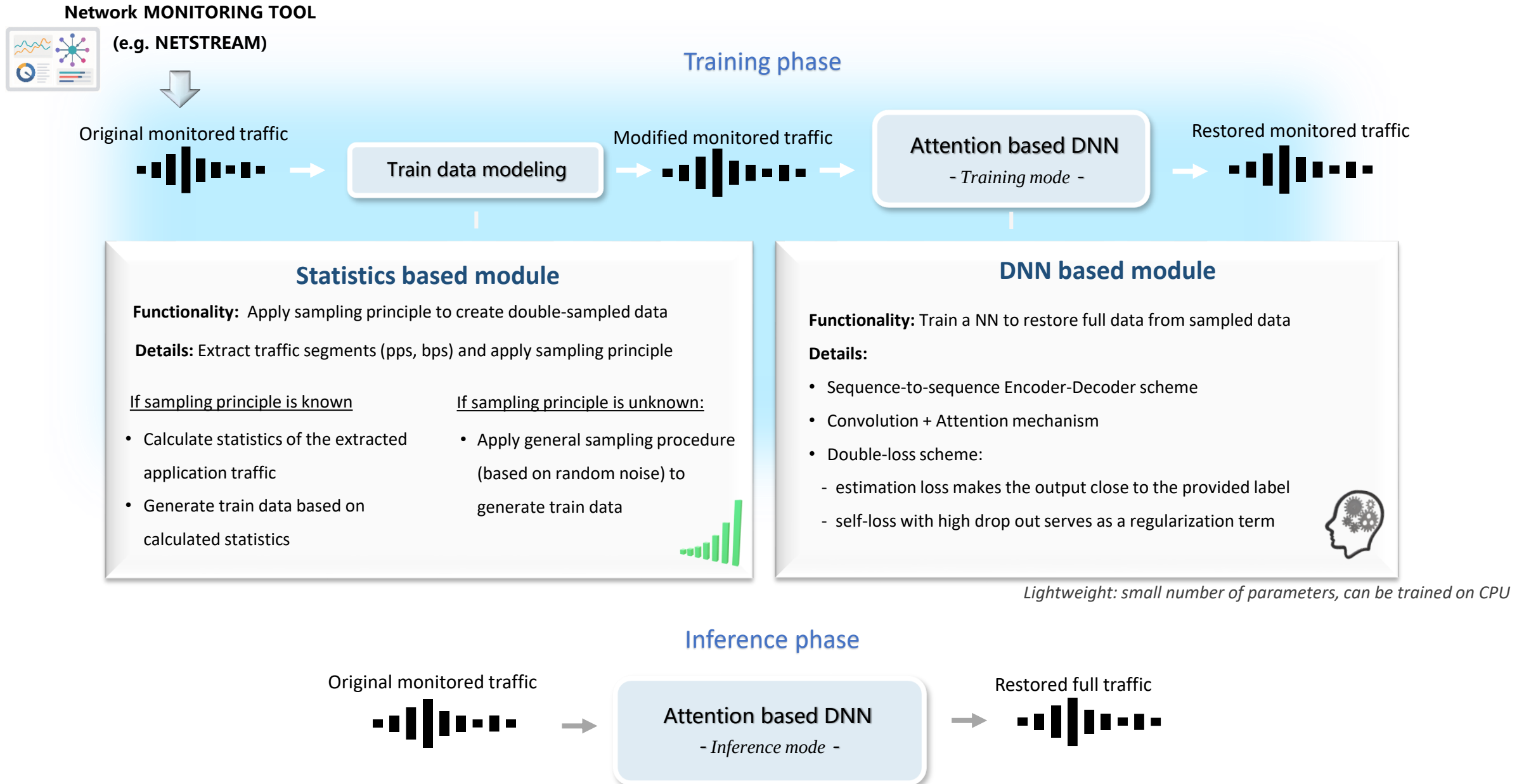
(a) with random sampling



(b) with Poisson sampling

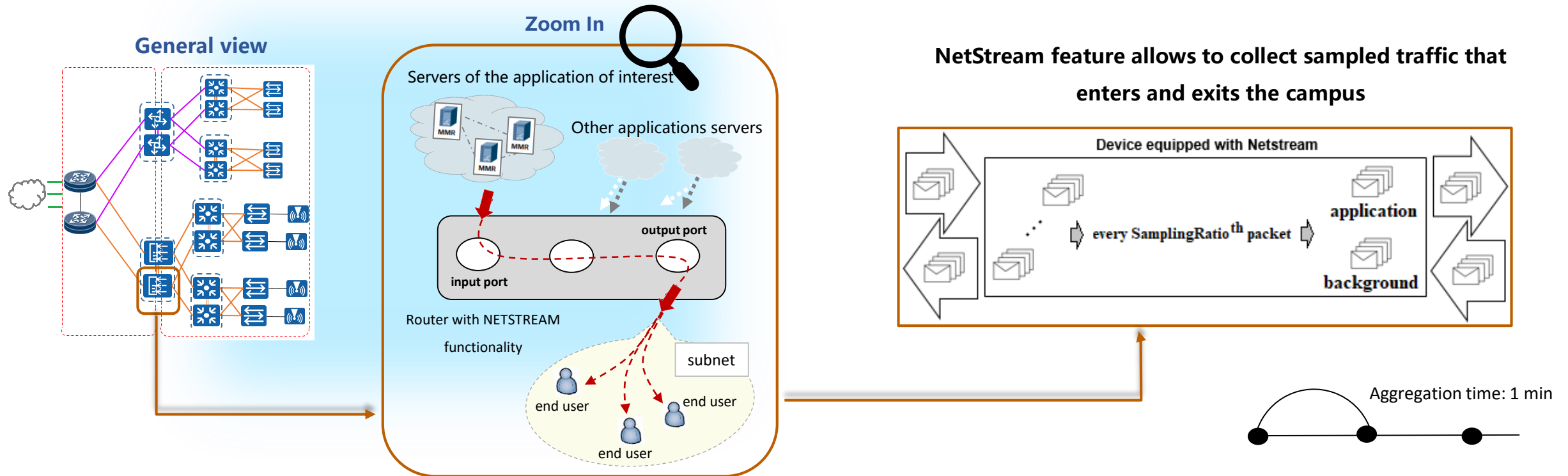
Conclusion: Although the network trained on double-sampled data, that underwent random sampling (instead of Poisson one) succeeded to restore the overall pattern, its result is much noisier and some important peaks are missed or underestimated

Model structure



Use case: Campus Network

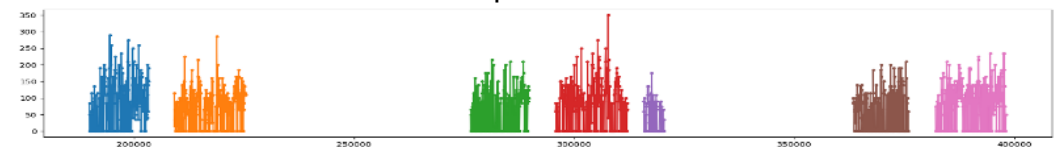
Real campus network with 2 routers equipped with NetStream feature are mounted on top of the campus network



Requirement:

To reconstruct the complete traffic generated by a Conference Call (CC) application on a per-user basis

A few examples of the CC traffic.



Each color denotes a separate conference call PPS along the time axis

Conference Call traffic inversion

Training

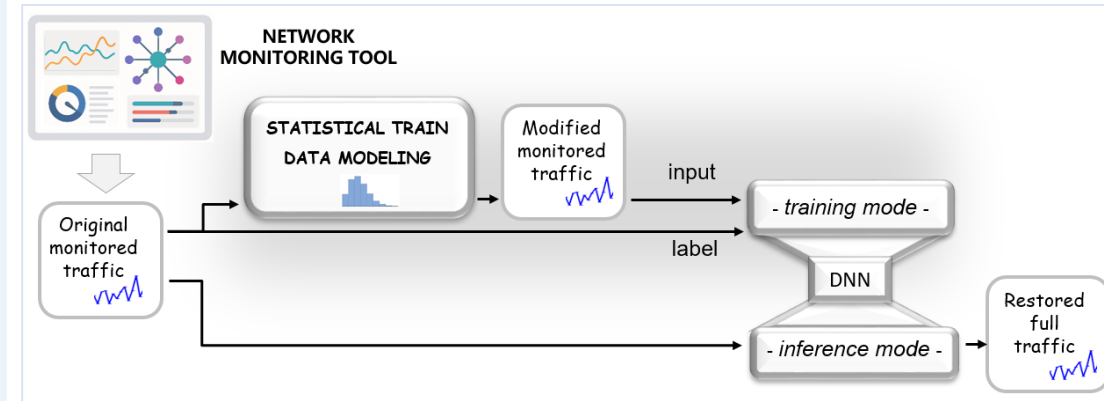
Sampled data: NetStream measurements contained 19 large batches, each batch including both CC and background traffic samples from about 5-7 days of records

Data preparation

- 1) Define input CC traffic by identifying CC server IP ('source IP' from data tuple)
- 2) Extract CC PPS and BPS segments corresponding to various users (users = different 'destination IP' from data tuple)

Model training

- 1) Use extracted CC segments as inputs to the model
- 2) Train model



Verification

Sampled data: NetStream measurements.

Ground truth: a few samples with total duration of 1,584 min of full fine grained CC traffic records (obtained with Wireshark).

Mean Absolute Percentage Error

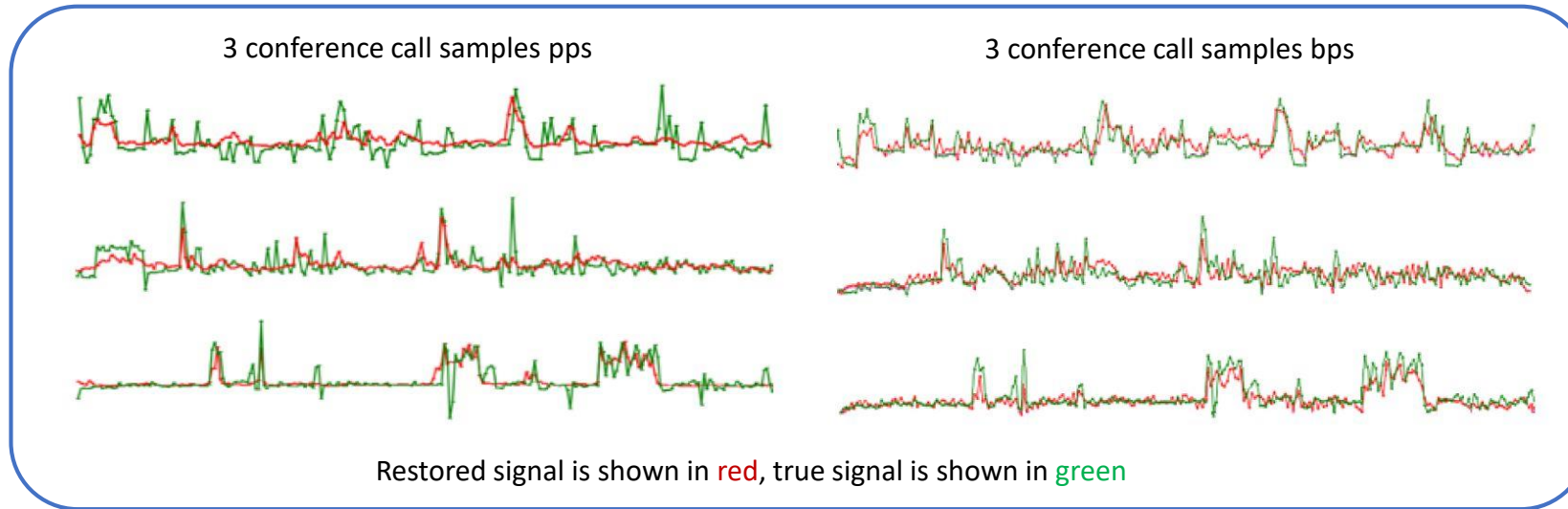
$$MAPE = \frac{100\%}{n} \sum \frac{|True(t) - Estimation(t)|}{|True(t)|}$$

Symmetric Mean Absolute Percentage Error

$$SMAPE = \frac{100\%}{n} \sum \frac{|True(t) - Estimation(t)|}{0.5 * |True(t) + Estimation(t)|}$$

Average inference time for 50 timestamp-long chunk: 17 ms for max number of parameters required. Can be further finetuned

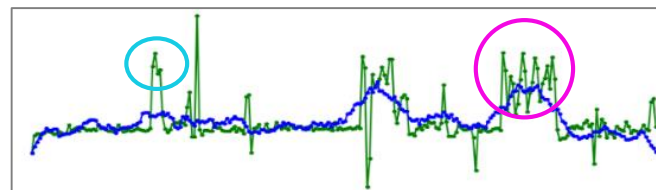
Evaluation and results



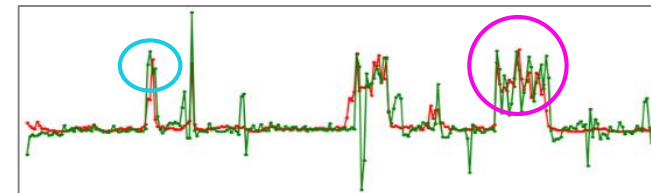
Method	MAPE (pps)	SMAPE (pps)	SMAPE (bps)
Upscaling (unsmoothed)	31.4%	14.0%	33.8%
Upscaling (smoothed)	18.0%	11.5%	27.3%
Our method (DNN + stats)	11.7%	5.6%	13.1%

Visual comparison of the statistics-based method (blue) and our method (red) to the ground truth (green).

Statistical results hardly restore peaks (cyan circles) and over smooth the data (magenta circles)



SMAPE = 13%

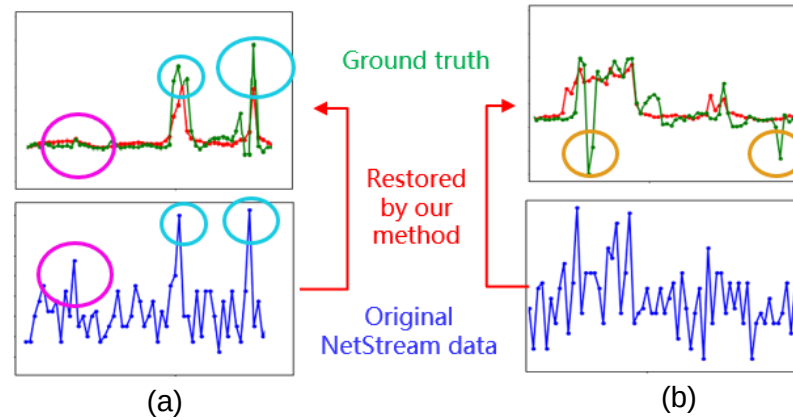


SMAPE = 5.1%

Summary and Discussion

Developed a **self-supervised algorithm** to restore full traffic data from highly sampled traffic measurements

- No labeled training data required
- Low cost (adjustable number of parameters, can be trained on CPU, short inference time)
- Efficient & Accurate



(a) Method restores true peaks (cyan circles) and ignores irrelevant sampled peaks (magenta circle)

(b) Limitations: Low values are not presented in NetStream data and cannot be restored (yellow circles)

General applicability: Due to the standard input the product can be applied to the data collected by other vendors tools (like NetFlow)

Additional benefits and suggestions

- **Cost reduction:** A trade-off between sampling ratio and inversion accuracy can be found.
- **Optimization:** Internal parameters (aging cache etc.) can be fine-tuned with respect to the desired accuracy.



Thank you